

- 25 A mixture of 1 mol of chlorine and 2 mol of nitrogen monoxide is allowed to come to equilibrium at a given temperature. The equilibrium mixture contains x mol of chlorine.



What are the amounts of the other constituents of the equilibrium mixture?

	NO / mol	NOCl / mol
A	$2x$	$2x$
B	$2 - 2x$	$2x$
C	$2x$	$2 - 2x$
D	$2 - x$	$2 - x$

- 26 Halothane, CF_3CHBrCl , is an anaesthetic.

How many constitutional isomers, including halothane, are there of molecular formula $\text{C}_2\text{HBrClF}_3$?

- A 3 B 4 C 7 D 8

- 27 P and Q are two straight-chain unsaturated hydrocarbons with the molecular formula C_4H_8 . Each react separately with bromine gas. The products of the two reactions are different isomers.

Which pair of names could identify the two organic **products** made?

- A 1,1-dibromobutane and 1,2-dibromobutane
B 1,2-dibromobutane and 2,3-dibromobutane
C 1,3-dibromobutane and 2,4-dibromobutane
D 2,3-dibromobutane and 3,3-dibromobutane